

Java Assignment - Student Management System

Objective

Write a Java program that simulates a simple student management system using arrays, functions, classes, objects, constructors, loops, conditional statements, and switch-case.

Program Requirements

Create a class **Student** that stores the student's ID, name, and grade.

Ask the user to enter **N**, where N represents the number of students.

Create the array based on the entered value:

```
Student[] students = new Student[n];
```

Read the information for all students from the user.

Main Menu

1. Display Students
2. Calculate Average Grade
3. Find Highest Grade
4. Search Student by ID
0. Exit

Features

- Display all students with their grade status.

Hint: Consider creating a method inside the **Student** class that displays the student's information instead of writing all the display code in the main method.

Grade Status

Grade ≥ 90 : Excellent

Grade ≥ 75 : Very Good

Grade ≥ 60 : Pass

Otherwise : Fail

- Search for a student using ID.
- Calculate the average grade.
- Find the student with the highest grade.

Input Validation

Validate that grades are between 0 and 100. If an ID is not found, display "Student not found."

Bonus Challenges

- Count passed students and failed students.
- Sort students by grade.
- **Each student has two subjects instead of one.**

Store the two subject grades in an array of size **2** inside the Student class.

The student's final grade should be calculated as the average of the two subject grades.

Use the final grade when displaying the student's status, calculating the class average, and finding the highest-grade student.

Note

Design your program using object-oriented programming principles. Try to keep each class responsible for its own tasks.